#This code will be used to demonstrate theorem 6.18 • We will demonstrate that Q2 $= \dfrac{((w+z)^3-y^3)\cdot x\cdot(-w^2\cdot(z^3-y^3)+z^2\cdot(w^3-y^3))}{y\cdot(w+z)^2\cdot((x^3-w^3)\cdot(z^3-y^3)+(x^3-z^3)\cdot(w^3-y^3))}$ ***is increasing along the γ2 curve; (Part 3 − calculating the sign of the derivative of Q2'(γ2 (1), 1))***

#Define Q2;

$$Q2 := \frac{((w+z)^3-y^3)\cdot x\cdot(-w^2\cdot(z^3-y^3)+z^2\cdot(w^3-y^3))}{y\cdot(w+z)^2\cdot((x^3-w^3)\cdot(z^3-y^3)+(x^3-z^3)\cdot(w^3-y^3))}:$$

#Apply the rational parametrization constructed in this chapter 4.

$$parametrization := \left\{ \mathrm{w}=\mathrm{a}+\frac{1}{a},\right.$$

$$z=b+\frac{1}{b},$$

$$x=a-\frac{1}{a}+b-\frac{1}{b},$$

$$\left. y=a-\frac{1}{a}-b+\frac{1}{b}\right\}:$$

$q2 := simplify(\ (\mathrm{expand}(\mathrm{subs}(\mathrm{parametrization}, \mathrm{Q2})))\ );$

$$q2 := \left(\left(a^2+\frac{b^2}{3}\right) b\, (-a^4 b^8 + 2\,(a^5-2\,a^3)\,b^7 + (a^6+10\,a^4-5\,a^2)\,b^6 + 2\,(-a^7+6\,a^3 - 2\,a)\,b^5 + (a^8-6\,a^6+4\,a^4+10\,a^2-1)\,b^4 + 2\,(2\,a^7-6\,a^5+a)\,b^3 + (7\,a^6-6\,a^4 + a^2)\,b^2 + 2\,(2\,a^5-a^3)\,b + a^4)\,(a\,b-1)^2\right) \Bigg/ \left(2\left(\frac{a^6 b^{12}}{6} + \frac{(a^7-a^5)\,b^{11}}{2} + \frac{(-a^8-a^4)\,b^{10}}{2} - \frac{\left(a^4+a^2+\frac{2}{3}\right) a^3 b^9}{2} + a^4\left(-1+a^6+a^4+\frac{3}{2}\,a^2\right) b^8 + (-4\,a^7 + a^5-a^3)\,b^7 + \left(-\frac{1}{3}+a^2-a^{10}-4\,a^8+a^4\right) b^6 + (-4\,a^7+a^5-a^3)\,b^5 + a^4\left(-1+a^6 + a^4+\frac{3}{2}\,a^2\right) b^4 - \frac{\left(a^4+a^2+\frac{2}{3}\right) a^3 b^3}{2} + \frac{(-a^8-a^4)\,b^2}{2} + \frac{(a^7-a^5)\,b}{2} + \frac{a^6}{6}\right)(a + b)\right) \tag{1}$$

#Define r2

$$r2 := \Bigg(-1024\left(b^2-\frac{1}{2}\right) b^{13}\,(b^2-5)\,(b^2+1)^4\,a^{23} + a^{32}\,b^{16} + (-4\,b^{18}+8\,b^{16}-4\,b^{14})\,a^2 + (\ldots$$

$$-4 b^{18} + 8 b^{16} - 4 b^{14}) a^{30} + (6 b^{20} - 32 b^{18} + 44 b^{16} - 32 b^{14} + 6 b^{12}) a^{28} + b^{16} - 9 \Big(b^{16}$$

$$+ \frac{4168}{9} b^{14} - 528 b^{12} - 328 b^{10} + \frac{3698}{9} b^{8} - 72 b^{6} - 16 b^{4} + 8 b^{2} + 1\Big) b^{8} a^{24} - 4 b^{2} (b^{28}$$

$$+ 12 b^{26} + 37 b^{24} + 378 b^{22} + 4187 b^{20} + 19656 b^{18} + 41514 b^{16} + 57358 b^{14} + 65514 b^{12}$$

$$+ 34120 b^{10} + 4315 b^{8} + 1018 b^{6} - 155 b^{4} + 12 b^{2} + 1) a^{14} - 4 b^{2} (b^{28} + 76 b^{26} - 539 b^{24}$$

$$- 2118 b^{22} + 5211 b^{20} + 36360 b^{18} + 92394 b^{16} + 78414 b^{14} + 38058 b^{12} + 17992 b^{10} - 2085 b^{8}$$

$$- 518 b^{6} + 37 b^{4} + 12 b^{2} + 1) a^{18} + 8 \Big(b^{20} + b^{18} - 699 b^{16} + 2369 b^{14} - \frac{1037}{2} b^{12} - 1827 b^{10}$$

$$+ \frac{2163}{2} b^{8} - 159 b^{6} - 27 b^{4} + b^{2} + 1\Big) b^{6} a^{22} - 1024 b^{9} (b^{2} + 1)^{4} a^{7} + 8 b^{6} \Big(b^{20} + b^{18}$$

$$- 27 b^{16} + 33 b^{14} - \frac{2189}{2} b^{12} - 3875 b^{10} - \frac{9357}{2} b^{8} - 3231 b^{6} - 2747 b^{4} - 1055 b^{2} - 127\Big)$$

$$a^{10} + 2 b^{4} (b^{24} + 36 b^{22} + 180 b^{20} - 1372 b^{18} - 9036 b^{16} - 25400 b^{14} - 54798 b^{12} - 64184 b^{10}$$

$$- 27212 b^{8} - 6620 b^{6} - 2636 b^{4} + 36 b^{2} + 1) a^{12} + (-1024 b^{22} + 72 b^{20} + 692 b^{18} + 8 b^{16}$$

$$- 76 b^{14} + 72 b^{12}) a^{26} + (-254 b^{28} + 840 b^{26} + 6504 b^{24} - 696 b^{22} - 25752 b^{20} - 169584 b^{18}$$

$$- 181788 b^{16} - 85104 b^{14} - 50328 b^{12} - 4792 b^{10} + 360 b^{8} + 72 b^{6} + 2 b^{4}) a^{20} + (b^{32} + 8 b^{30}$$

$$+ 20 b^{28} + 1672 b^{26} + 1950 b^{24} - 32376 b^{22} - 157780 b^{20} - 317080 b^{18} - 279672 b^{16}$$

$$- 222104 b^{14} - 125524 b^{12} - 7288 b^{10} - 3426 b^{8} - 120 b^{6} + 20 b^{4} + 8 b^{2} + 1) a^{16} + (-9 b^{24}$$

$$- 72 b^{22} + 144 b^{20} - 1912 b^{18} + 142 b^{16} + 4232 b^{14} + 2192 b^{12} - 1864 b^{10} - 265 b^{8}) a^{8}$$

$$+ (72 b^{20} - 76 b^{18} + 1288 b^{16} + 692 b^{14} - 696 b^{12} - 512 b^{10}) a^{6} + (6 b^{20} - 32 b^{18} + 44 b^{16}$$

$$- 288 b^{14} - 250 b^{12}) a^{4} - 2560 \Big(b^{6} - \frac{38}{5} b^{4} + \frac{42}{5} b^{2} - \frac{11}{5}\Big) b^{11} (b^{2} + 1)^{4} a^{21}$$

$$- 1536 b^{9} \Big(b^{8} - \frac{43}{3} b^{6} + \frac{95}{3} b^{4} - \frac{52}{3} b^{2} + \frac{7}{3}\Big) (b^{2} + 1)^{4} a^{19} + 512 b^{7} (b^{10} + 11 b^{8} - 78 b^{6}$$

$$+ 81 b^{4} - 22 b^{2} + 1) (b^{2} + 1)^{4} a^{17} + 512 b^{7} (b^{10} - 10 b^{8} - 9 b^{6} + 42 b^{4} - 19 b^{2} + 1) (b^{2}$$

$$+ 1)^{4} a^{15} - 2560 \Big(b^{8} - \frac{16}{5} b^{6} + b^{4} - \frac{3}{5} b^{2} + \frac{3}{5}\Big) b^{7} (b^{2} + 1)^{4} a^{13} + 2560 \Big(b^{6} - \frac{6}{5} b^{4}$$

$$+ \frac{8}{5} b^{2} - 1\Big) b^{7} (b^{2} + 1)^{4} a^{11} + 512 b^{7} (b^{2} - 2) (b^{2} + 1)^{5} a^{9}\Big) :$$

#Calculating the derivative of Q2 at b=1

$$d := eval(diff(q2, a), b=1) \cdot \left(-\frac{eval(diff(r2, b), b=1)}{eval(diff(r2, a), b=1)} \right) + eval(diff(q2, b), b=1) :$$

#Locate a range for $\gamma 2(1)$

$sturm(sturmseq(eval(\ r2\ , b=1), a)\ , a, 2.6, 3)$; *#number of roots in [2.6,3)*

$$1 \tag{2}$$

#Define nd

$nd := numer(d)$

$$nd := (a-1)\,(9\,a^{53} - 81\,a^{52} - 96\,a^{51} + 540\,a^{50} + 2445\,a^{49} + 891\,a^{48} - 3360\,a^{47} + 16336\,a^{46} - 6746\,a^{45} - 205830\,a^{44} - 3271872\,a^{43} - 1580968\,a^{42} + 17978998\,a^{41} + 111198698\,a^{40} + 54640096\,a^{39} - 253761712\,a^{38} - 1126401941\,a^{37} - 686298067\,a^{36} - 325640672\,a^{35} + 1309779844\,a^{34} - 58731873\,a^{33} + 3713940105\,a^{32} - 1214895680\,a^{31} + 4099046048\,a^{30} - 3424070620\,a^{29} + 3279665628\,a^{28} - 4846574976\,a^{27} + 2103950928\,a^{26} - 3356527900\,a^{25} + 1412097308\,a^{24} - 2121490752\,a^{23} + 1001092640\,a^{22} - 512079601\,a^{21} + 518605145\,a^{20} - 180857632\,a^{19} + 250200292\,a^{18} + 107574459\,a^{17} + 42795293\,a^{16} + 45674336\,a^{15} + 10753936\,a^{14} + 31192918\,a^{13} - 3329782\,a^{12} + 2365504\,a^{11} + 3331800\,a^{10} - 432698\,a^{9} + 137690\,a^{8} + 120672\,a^{7} - 3952\,a^{6} + 3421\,a^{5} + 1003\,a^{4} - 160\,a^{3} - 4\,a^{2} - 7\,a - 1) \tag{3}$$

#Define dd

$dd := denom(d)$

$$dd := 4\,(3\,a^{10} - 9\,a^{8} - 24\,a^{7} + 10\,a^{6} - 6\,a^{4} - 8\,a^{3} + 3\,a^{2} - 1)^{2}\,(a+1)^{2}\,a^{3}\,(a^{28} - 7\,a^{24} - 208\,a^{22} + 405\,a^{20} + 23552\,a^{19} + 1232\,a^{18} + 10752\,a^{17} - 319075\,a^{16} - 48640\,a^{15} - 592416\,a^{14} - 26112\,a^{13} - 570845\,a^{12} + 23040\,a^{11} - 398944\,a^{10} + 19968\,a^{9} - 143253\,a^{8} + 5632\,a^{7} - 42000\,a^{6} - 4608\,a^{5} + 647\,a^{4} - 3584\,a^{3} + 144\,a^{2} - 65) \tag{4}$$

#Verify that nd and dd have the same sign in the interval.

$sturm(sturmseq(\mathrm{nd}, a)\ , a, 2.6, 3)$; *#number of roots in [2.6,3) ;*

$sturm(sturmseq(\mathrm{dd}, a)\ , a, 2.6, 3)$; *#number of roots in [2.6,3)*

$$0$$

$$0 \tag{5}$$

#Calculate the sign of $Q2'\,(\gamma 2\,(1), 1)$ using $Q2'\,(3, 1)$

$$\frac{eval(nd, a=3)}{eval(dd, a=3)}$$

$$\frac{67443961872063043}{22995239986470285} \tag{6}$$